LEF MODEL · CHAPTER 21

Chapter 21: The Law of Moons in the Solar System — the Ruler of Birth Modes, the Pump Principle and the Atmospheric Lid

Based on Paper 11 — 11. Moon Births: One mechanism for the births of moons (Parts A–B: the birth ruler, the pump and lid principles, the stages of differentiation and the protocol of 13 checks)

01 · THE MYSTERY

The mystery: does one physical law explain the birth of all the moons in the solar system — from the Earth to Pluto?

In the previous chapters we saw how the Earth created the Moon on the tip of a pear (Chapters 1–9), and how Mars ejected Phobos and Deimos from a volcano at the top of a pointed bulge (Chapters 19–20). In Paper 11 the LEF model generalises these two cases into **"the Moon Law"** — a uniform mechanism valid for every planet and dwarf planet in the solar system:

1. **Why are the moons so different in size and composition?** The Earth's Moon is dry and huge relative to its parent (1.2% of the Earth's mass); the moons of the giants (Jupiter, Saturn, Uranus and Neptune) are tiny (0.01%–0.02%) and rich in water and ice; while Charon makes up 11% of the mass of the Pluto–Charon system!

2. **Why does the density of the moons fall with distance from the Sun — and then jump again at Uranus?** The density of the Earth's Moon is 3,344 kg/m³,

Jupiter's Galilean moons ~ 2,600 kg/m³, Saturn's moons ~ 1,350 kg/m³ — but at the more distant Uranus the density rises back to 1,505 kg/m³!

3. **Why in five planets (Mars, Jupiter, Saturn, Uranus and Neptune) do small moons sit on both sides of the corotation line?** And why were Mercury and Venus left entirely without moons?

02 · AN IMAGE

An image: the distance ruler of the corotation line, a skater pulling in her arms, and two pots of soup with lids of foam

Four tangible images explain the physics of the birth of moons across the whole solar system:

- **The ice skater pulling her arms in while her coat puffs outward (shrinking inside and swelling outside, A.5):** how does planetary differentiation bring the corotation line close to the surface? A body that swells uniformly without losing angular momentum would slow down and push the corotation line away. But differentiation is an opposite process inside and outside: the heavy core (holding most of the mass) **sinks and shrinks inward** and speeds up the rotation like a skater pulling in her arms, while the light water and volatiles are pushed upward and **swell the outer envelope** almost without slowing the rotation. So the relative distance of the corotation line shrinks according to $x \propto (k^2 R)^{1/3}$ and meets the surface!
- **"The hydrodynamic pump" (The Pump Principle, A.4):** the sinking of the heavy materials into the core pushes, by hydrostatic pressure, water and light silicates upward to the ejection openings (the tip of the pear or the vent of the volcano on the bulge). As long as the pump is active and the crust is not sealed, **the Birth Window** stays open.
- **Two pots of soup on the fire — a thick lid of foam against a thin lid of foam (the atmospheric lid principle, A.4 & A.9):** the distance from the Sun determined how much hydrogen and helium each giant captured before the protoplanetary disc dispersed. Saturn (9.5 AU) has **a thick gas envelope**

(70%–85% of the mass) that acts as a thermodynamic lid over the water: the vapours condense and fall back inward, the upper layer stays saturated with water and soft, the corotation line stays in place (1.92 R), and the ejected moons are light and rich in ice (1,346 kg/m³). Uranus (19 AU) has **a thin gas lid (10%–20%)**: the vapours escaped to space through the corotation line, carried away angular momentum that pushed the line out to 3.24 R, and dried and compacted the upper ejection layer — which is why the moons of Uranus are denser (1,505 kg/m³)!

- **The three birth modes across the solar system (A.4, L.4):** at the hot, dry inner edge (the Earth) a large, dry moon was built on the tip of a pear (**Built**); in the wet central zone (from Mars to Neptune) small, water-rich moons were ejected from volcanoes on bulges (**Ejected**); and at the cold, fast outer edge (Pluto–Charon) a unified body was stretched into a "peanut" and torn at its waist (**Split**).

03 · SCIENCE AND MODEL

Science and model: the birth law, the ruler of states and the four stages of differentiation (Part A)

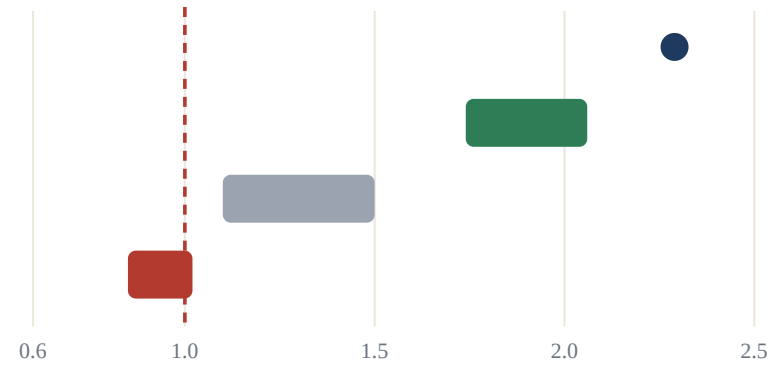
Part A of Paper 11 formulates the precise quantitative conditions of the Moon Law:

1. **The Ruler of Birth Modes (A.3):** if the radius of the corotation line is measured in units of the parent's radius ($x = a_c/R$), the parent's radius cancels out completely! The position x depends **only on the ratio between the rotation rate and the mean density** ($x^3 = 4\pi G\rho/3\Omega^2$), so one ruler applies to every body in the universe:
    - 2 **Pear (Built on Pear Tip):** x ≈ 2.29 (the Earth at 5.77 hours and a density of 3,940 kg/m³ — the Moon makes up 1.2% of the parent's mass).
    - 3 **Bulge and volcano (Ejected on Bulge):** x ≈ 1.74–2.06 (Mars at 7–9 hours and a density of 1,170 kg/m³, and likewise the gas and ice giants — tiny moons of 0.01%–0.02%).
    - 4 **A volcano without a bulge (a theoretical possibility):** x ≈ 1.1–1.5.

- 5 **A full peanut split (Split at Neck):** x ≈ 0.85–1.02 (passing the elongation threshold x = 1.53 to the tearing threshold x = 1 — Pluto and Charon at 1.9–2.5 hours, Charon 11% of the pair's mass).

6 **The composition of the moon as a chronological cross-section of the parent's stages of differentiation (A.9):** as differentiation progresses, the corotation line moves inward (at Jupiter and early in Saturn's life), so **more distant moons were born earlier** and closer moons were born later:

- 7 **Stage 0 (the start of differentiation — centrifugal spraying of water):** spraying of primordial water from the outer layer above the Roche limit → airy ice moons (Iapetus 1,088 kg/m³, Hyperion 544 kg/m³).
- 8 **Stage 1 (early — mixed material):** ejection of material representing the composition of the whole parent before the sorting was complete → large moons with a density similar to the parent's inner body (f ≈ 0.6–0.7: Callisto 1,834, Ganymede 1,936, Titan 1,880, Titania 1,711, Oberon 1,630 kg/m³).
- 9 **Stage 2 (the peak of the hydrodynamic pump):** pushing of magma saturated with water and light silicates; the escape of volatiles to space leaves high porosity ("sponginess", 30%–40%) and low density → Phobos (1,860), Deimos (1,470), Tethys (984), Miranda (1,200 kg/m³).
- 10 **Stage 3 (the end of differentiation):** ejection of a dry, dense silicate residue with a defined metallic core → Io (3,528 kg/m³).



Built on a pear tip · 2.29

Ejected on a bulge · 1.74–2.06

Volcano without a bulge · 1.1–1.5

Peanut split · 0.85–1.02

Position of the corotation line in parent radii · red line: the tearing threshold

The ruler of birth modes: the same number x determines whether a moon was built on a tip, ejected from a volcano or torn from a peanut — for every body, regardless of the parent's radius.

The right density comparison – isolating the inner body from the gas envelope (A.9)

Moons are born from the inner body (rock and ice) and not from the hydrogen and helium envelope. When the mean density of the moons is split into a silicate-metal component (3,500 kg/m³) and an ice component (950 kg/m³), the silicate mass fraction f falls gradually from the Earth (f = 0.98, $\rho = 3,344$), through Jupiter (f = 0.83, $\rho = 2,578$) to Saturn (f = 0.36, $\rho = 1,346$) — and rises back at Uranus (f = 0.49, $\rho = 1,505$), because Uranus' thin gas lid (10%–20%) let the water escape and dried the upper ejection layer!

04 · THE MECHANISM IN WORDS

The mechanism in words: the sixteen clauses of the Moon Law (A.10) and the protocol of 13 checks (Part B)

Parts A.10 and B of Paper 11 unite all the physical conditions into one system of laws and one uniform measurement protocol of 13 checks for every body in the solar system:

1. **The Sun as a thermodynamic clock — the arc of births (A.10.4 & L.4):** the birth of moons requires at the same time active differentiation, an open time window, an unsealed crust and a subsurface layer of low viscosity. The distance from the Sun determines how long the four conditions hold together: at the inner edge (Mercury and Venus) fast drying closes the birth window early, and only the Earth managed to build one moon on the tip of a pear; in the central zone (from Mars to Saturn) moons are ejected in abundance up to the peak at Saturn (~ 24 regular moons); and at the cold outer edge (Uranus, Neptune and Pluto) the freezing of the crust gradually blocks the openings of the pump and shifts the system to mechanical splitting.

2. **The structural race and the feedback of the moon's mass (A.4, A.10.16):** for a moon to be born, the radial growth rate of the pear or the bulge must exceed the rate at which the corotation line recedes. After the birth, if the mass ratio of the moon to the parent is large (like the Earth's Moon, 1.2%), the moon carries most of the angular momentum (84%) and slows the parent; but in the giants and in Mars the mass of the moons is tiny (0.01%–0.02%, only 0.4%–1.4% of the momentum), so any retreat of the corotation line comes **only from the escape of water and gas through the line ($\Delta L_{\text{escaped}}$)**!

3. **The systematic protocol of 13 checks (Part B):** to avoid bias, every planet and dwarf planet is checked by 13 fixed parameters in chronological sequence: (1) global basic data; (2) catalogue and separation of regular from captured moons; (3) dynamic reconstruction of the ancient rotation (check 3 — using the radius and the moment-of-inertia coefficient k at the time of birth!); (4) verification of the birth law; (5) analysis of water and volatiles; (6) the degree of differentiation today ($k = C/MR^2$); (7) definition of the birth window; (8) the path of tidal migration (check 8 — integrating da/dt to identify the need for a secondary push from a ring); (9) survival parameters (Roche limit and Hill radius); (10) morphological traces on the parent; (11) structural traces on the moons (porosity, tidal locking and a frozen bulge); (12) analysis of ring systems as a continuous flow; and (13) isolation of anomalies.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The position of the corotation line in units of the parent's radius ($x = a_c/R$), the birth day (P_{birth}), and the effect of the inner shrinking and the outer swelling during differentiation are formulated as follows (§A.3, §A.5, §B):

$$x^3 = \frac{4\pi G\rho}{3\Omega^2}$$

$$P_{\mathrm{birth}} = \sqrt{\frac{3\pi x^3}{G\rho}}$$

$$P_{\mathrm{critical}} = \sqrt{\frac{3\pi}{G\rho}} \quad (x = 1)$$

$$L = kMR^2\Omega = \mathrm{const} \implies x \propto (k^2 R)^{1/3}$$

$$r_H = a\left(\frac{M}{3M_\odot}\right)^{1/3}$$

Since the moment-of-inertia coefficient k appears squared in the differentiation formula $x \propto (k^2 R)^{1/3}$, a drop in k from 0.40 (a homogeneous body) to 0.20–0.25 (a compact core) brings the corotation line x from 2.10 R to 1.44–1.68 R even when the outer envelope swells by 1.3 times — and this is the geodynamic engine that brings the surface and the corotation line together!

06 · THE SIGNATURE

The signature in reality: reconstructing the corotation line and the mixing equation of the moons

Applying checks 3 and 5 of the protocol to the planets reveals an impressive quantitative match between the chemical composition of the moons and the

conservation of angular momentum (§A.9, §B):

- **The two-component mixing equation of the moons' density (§A.9):** by the formula $\frac{1}{\rho} = \frac{f}{3500} + \frac{1-f}{950}$, the silicate mass fraction f of the large moons born in Stage 1 of differentiation converges to a narrow, uniform range of **f ≈ 0.6–0.7** across the whole outer solar system (Callisto, Ganymede, Titan, Titania, Oberon, and also the Pluto–Charon system)!
- **The test of reconstructing the corotation line without mass loss ($\Delta L_{\text{escaped}} = 0$, Part B):** when only the angular momentum of the moons is returned to the parent, at Jupiter (2.29 R → 2.28 R) and Saturn (1.92 R → 1.90 R) the line hardly moves (consistent with the thick-lid principle); at the Earth the line returns to 2.29 R (since the Moon carries 84% of the momentum); while at Mars (6.03 R), Uranus (3.24 R) and Neptune (3.39 R) the line stays far from the birth range (x ≈ 2) — proof that in all three, enormous amounts of water and gas escaped through the corotation line!

The lab box: the ruler of birth modes, the stages of differentiation, the two-pot model and the protocol of 13 checks (Paper 11, Parts A–B)

Birth mode	Mechanism lifting the material	Position of the corotation line ($x = a_c/R$)	Moon/parent mass ratio	Zone in the solar system and example
1. Built on a tip	**Prolonged accumulation at the tip of a planetary pear; detachment on crossing the line**	**2.29 (5.77 h, 3,940 kg/m³)**	**1.2% (large and dry)**	**The dry inner edge: the Earth's Moon (a single case)**
2. Ejected on a bulge	**An equatorial bulge and a volcano injecting magma saturated with water**	**1.74–2.06 (Mars 7–9 h, 1,170 kg/m³)**	**0.01%–0.02% in the giants (1.9 × 10^{-8} at Mars)**	**The wet central zone: Phobos, Deimos and the moons of the giants**
2b. A volcano without a bulge	Point ejection from a local volcano	≈ 1.1–1.5	Tiny	A theoretical possibility (no calculated example)

3. Full split	**Rotation beyond the fission threshold (x = 1); elongation into a peanut and tearing at the waist**	**0.85–1.02 (1.9–2.5 h, 1,840 kg/m³)**	**11%–12% (double locking)**	**The cold outer edge: Pluto and Charon (and Haumea before splitting)**

Stage of differentiation in the parent (A.9)	The geodynamic mechanism and the material at the opening	Properties of the moon formed	Examples in the solar system
Stage 0: the start of differentiation	Centrifugal spraying of water from the outer layer above the Roche limit	An airy, spongy ice moon of very low density	Iapetus (1,088), Hyperion (544 kg/m³)
Stage 1: early stage	**Mixed material representing the composition of the whole inner body**	**A large moon, density similar to the inner body (f ≈ 0.6–0.7), partial differentiation**	**Callisto (1,834), Ganymede (1,936), Titan (1,880), Titania (1,711), Oberon (1,630)**
Stage 2: the peak of the pump	**Fast core sinking pushing magma saturated with water and light silicates**	**The escape of volatiles creates "sponginess" (30%–40% porosity) and low density**	**Phobos (1,860), Deimos (1,470), Tethys (984), Miranda (1,200 kg/m³)**
Stage 3: the end of differentiation	Almost complete sorting; ejection of a dry, dense silicate residue	A dense silicate moon, completely dry and with a metallic core	Io (3,528 kg/m³)

07 · NEXT

The door ahead

In this chapter we formulated **the Moon Law** of the LEF model: the position of the corotation line in units of the parent's radius ($x = a_c/R$), the principle of the hydrodynamic pump, the principle of the atmospheric lid (the two-pot model of Saturn against Uranus), and the shrinking of the core that brings the corotation line to the surface.

In the next chapter — **Chapter 22: A Full Planetary Survey — from Mercury and Venus to Pluto, Haumea and Ceres** (from Paper 11 — Parts C–N) — we will run the

protocol of 13 checks planet by planet: we will discover why Mercury and Venus have no moons, how Saturn's rings work as a continuous flow in equilibrium, why Uranus is tilted by 98°, how Triton was captured as a peanut lobe from the Kuiper belt, and we will map the 12 open questions of the solar system!

Chapter 22

A Full Planetary Survey — from Mercury and Venus to Pluto, Haumea and Ceres →

← Chapter 20 · Prologue

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